

Node.js Lab: Memory Leak Diagnostics

Duration : 60 Min.

Scenario

An Express application endpoint `/api/track` accumulates client metadata in a global data structure without eviction, resulting in an active memory leak.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Identify the leaky array/collection pattern inside `server.js` and restrict cache sizing to a maximum threshold of 100 elements using Map eviction.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	File <code>server.js</code> exists in workspace	5 Marks
TC2	Leaking array resolved or replaced with size-limited collection	5 Marks
TC3	Express routes perform successfully under load	5 Marks
TC4	Memory heap consumption verified to remain bounded	5 Marks

Total Score: 20 Marks